



## **Does implied volatility efficiently predict future realized volatility?**

*And how does the answer differ between  
Bitcoin and the S&P 500?*

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### Does implied volatility predict future realized volatility?

*Same answer for Bitcoin and the S&P 500?*

- Implied volatility is the market's forecast of future risk.
- In equities, the relationship has been studied for thirty years.
- In crypto, it has barely been documented.

# What we know vs. what we don't

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## Equity markets

- Implied volatility is informative  
Christensen & Prabhala (1998)
- Slope below one, positive premium  
Bollerslev, Tauchen & Zhou (2009)
- Predictive power weakens in crises  
Whaley (2009), Giot (2005)

## Crypto markets

- DVOL exists only since March 2021
- No published predictive evidence
- No regime-conditional analysis of  
BTC option markets

*Our contribution: apply the equity benchmark to crypto, in a unified framework, with regime splits.*

## Five hypotheses

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- **H1.** Implied volatility overestimates future realized volatility on average: positive variance risk premium.
- **H2.** Implied volatility is informative but not a perfect predictor.
- **H3.** The variance risk premium is larger in crypto than in equities.
- **H4.** The predictive relationship weakens in stress periods.
- **H5.** Equity markets are close to efficient in normal periods; crypto is not.

# Data

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| Series               | Provider                 | Identifier    |
|----------------------|--------------------------|---------------|
| Bitcoin spot         | Binance public API       | BTCUSDT daily |
| Bitcoin implied vol. | Deribit public API       | BTC_DVOL      |
| S&P 500 close        | Yahoo Finance (yfinance) | GSPC          |
| VIX close            | Yahoo Finance (yfinance) | VIX           |

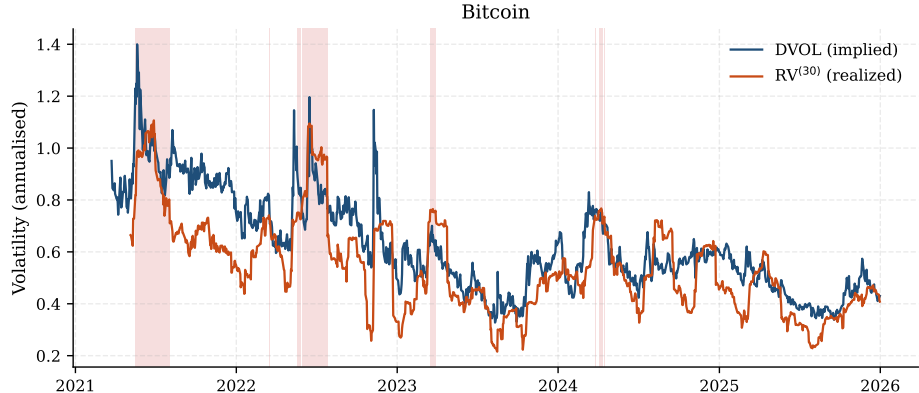
- **Period:** 24 March 2021 → 31 December 2025.
- **After NYSE-calendar inner join:** 1 201 daily observations.
- All series come from public APIs (no proprietary terminal data).

## Method, in one slide

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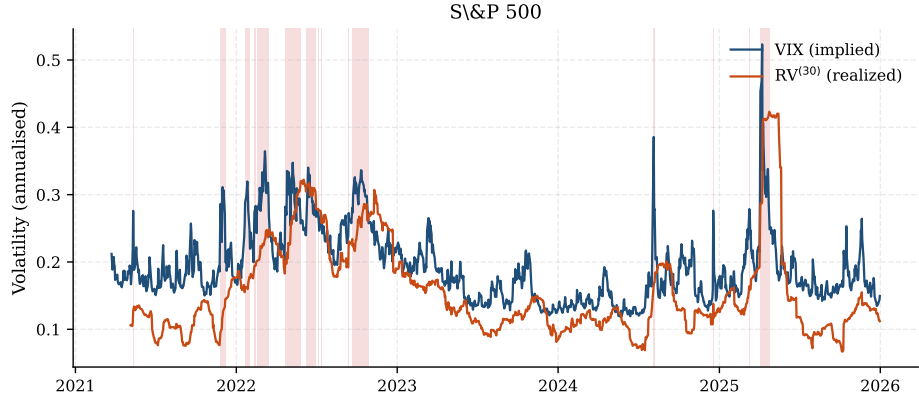
- **Realized volatility:** annualised standard deviation of 30-day log returns.
- **Predictive regression:** future realized volatility regressed on current implied volatility.  
*Slope = 1 means efficient.*
- **Regimes:** *stress* = top 10 % of VIX (equities) or BTC realized volatility (crypto), set ex ante.
- **Inference:** Newey–West HAC standard errors to correct for rolling-window overlap.

# A first look: Bitcoin



- DVOL and 30-day realized volatility move together, with a positive average wedge.
- Red shading: stress days (BTC 30-day RV above its 90th percentile, set ex ante).

## A first look: S&P 500



- Volatility runs in a much narrower band than Bitcoin, but spikes sharply in crises.
- Red shading: stress days (VIX above its 90th percentile, set ex ante).



## Implied volatility is informative, but biased

|                     | Bitcoin                  | S&P 500                 |
|---------------------|--------------------------|-------------------------|
| Slope $\hat{\beta}$ | <b>0.55***</b>           | <b>0.79***</b>          |
| HAC s.e.            | (0.09)                   | (0.10)                  |
| $R^2$               | 0.33                     | 0.36                    |
| $H_0 : \beta = 1$   | rejected ( $p < 0.001$ ) | rejected ( $p = 0.03$ ) |
| $N$                 | 1 171                    | 1 171                   |

- Both slopes are highly significant: implied volatility carries real information.
- Both are significantly below one: positive variance risk premium in both markets.

\*\*\* significant at 1 %. Standard errors: Newey–West HAC, 30 lags.

## The variance risk premium is larger in crypto

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*Crypto investors require more than twice the compensation per unit of volatility risk.*

Confirms H3.

## Headline result

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*In calm markets, the VIX looks almost efficient.*

$$\hat{\beta} = 0.97$$

S&P 500, normal regime,  $N = 955$

$$p = 0.89 \text{ for } H_0 : \beta = 1$$

**We cannot reject that the VIX is a fully efficient predictor  
of S&P 500 realized volatility in normal periods.**

## The premium reacts to stress in opposite directions

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|                    | Normal | Stress | Direction         |
|--------------------|--------|--------|-------------------|
| <b>Bitcoin</b>     | +0.087 | +0.055 | <i>compresses</i> |
| <b>S&amp;P 500</b> | +0.032 | +0.047 | <i>widens</i>     |

- **Equity stress**  $\Rightarrow$  premium *widens*. Implied volatility surges with hedging demand.
- **Crypto stress**  $\Rightarrow$  premium *compresses*. Realized volatility catches up faster than DVOL.

*An asymmetry only visible when both markets are studied together.*

## Why? Three economic mechanisms

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- 1 **Hedging demand.** Equity investors buy out-of-the-money puts in stress, which pushes the VIX above expected realized volatility.
- 2 **24/7 trading.** Bitcoin realized volatility adjusts to news within hours, not days, which compresses the wedge faster.
- 3 **Market maturity.** Equity option markets are deep and dominated by institutions; crypto options are concentrated on a few venues and a more retail base.

## What we don't claim

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- **Short sample.** DVOL only goes back to March 2021.
- **Aggregated indices.** No full option surface, so no skew and no term structure.
- **Daily frequency.** Intraday data would refine realized volatility.
- **Asymmetric regime definition.** VIX for equity stress vs. realized volatility for crypto stress.

## Take-aways

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- 1 Implied volatility is a **risk-adjusted expectation**, not a raw forecast, in both markets.
- 2 Equity markets are **close to efficient in normal periods**; crypto markets are **not**.
- 3 The variance risk premium **widens in equity stress** and **compresses in crypto stress**. This asymmetry is only visible across markets.

Code, data and full pipeline:  
[github.com/X9ClementX9/iv-rv-thesis](https://github.com/X9ClementX9/iv-rv-thesis)

# Thank you.

*Questions?*



**Replication materials**

`github.com/X9ClementX9/iv-rv-thesis`